Adaptive AI-Based Assessment Generation System

Problem Statement:

Educational assessments are generally static and identical for all learners, irrespective of their understanding, performance, or learning goals. Manual assessment creation is time-consuming and difficult to personalize for individual learners. Static assessments also fail to evaluate deeper conceptual understanding effectively. An adaptive AI-based assessment system can help automate question generation, improve learner engagement, and provide personalized evaluation experiences.

Recent advances in Large Language Models (LLMs) enable automatic generation of meaningful and context-aware questions from educational content. These models also make it possible to generate adaptive follow-up questions based on learner responses.

This project aims to develop a prototype system that can:

- Generate assessment questions from provided content or objectives
- Create follow-up questions dynamically based on learner responses
- Adapt question difficulty and context according to learner performance

The project will focus on building a functional prototype suitable for demonstration and evaluation within a two-month internship duration.

Weekly Plan for 2 Month Internship:

Week 1 - Onboarding and Project Understanding

- HR onboarding and internship formalities
- Understand project vision and expected outcomes
- Familiarization with tools, repositories, and workflows
- Study initial reading materials and reference implementations

Week 2 - Literature Review and Requirement Analysis

- Study research papers and existing adaptive assessment systems
- Review LLM-based question generation approaches
- Identify suitable question formats and adaptive strategies
- Finalize project requirements and scope

Week 3 - Question Generation Module

- Develop prompts for question generation
- Generate questions from input content/objectives
- Support basic question types and difficulty levels
- Evaluate generated question quality

Week 4 - Adaptive Follow-up Questioning

- Analyze learner responses
- Implement follow-up question generation logic
- Generate clarification and difficulty-based questions
- Improve contextual relevance of generated questions

Week 5 - Personalization and Assessment Flow

- Implement learner performance tracking
- Create adaptive difficulty selection logic
- Personalize assessments using prior interactions
- Build assessment flow management

Week 6 - Streamlit Prototype Development

- Develop Streamlit-based user interface
- Integrate question generation and adaptive workflow
- Display learner responses and generated feedback
- Enable interactive assessment sessions

Deliverables

- Literature review summary
- Workflow Architecture
- Working Streamlit prototype